MAT 221 CALCULUS I

REVIEW ON DERIVATIVES

Limit definition of derivative or first principle of derivative states:

Theorem: If $y=f(x)$ is a continuous function then

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

Example 1

Use limit definition to find $\frac{dy}{dx}$ of

(a) $y=\sqrt{x}+3$ (b) $y=\frac{2}{1-x}$

Solution

(a) Let $y=f(x)=\sqrt{x}+3$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\sqrt{x+h}+3-(\sqrt{x}+3)}{h}$$

$$= \lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} \cdot \frac{(\sqrt{x+h}+\sqrt{x})}{(\sqrt{x+h}+\sqrt{x})}$$

$$= \lim_{h\to 0} \frac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})}$$

$$= \lim_{h\to 0} \frac{1}{\sqrt{x+h}+\sqrt{x}} = \frac{1}{2\sqrt{x}}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

Solution

(b) Let $y = f(x) = \frac{2}{(1-x)}$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\frac{2}{1-(x+h)} - \frac{2}{1-x}}{h}$$

$$= \lim_{h\to 0} \frac{2(1-x) - 2[1-(x+h)]}{h[(1-x)(1-(x+h))]}$$

$$= \lim_{h\to 0} \frac{\cancel{2} - \cancel{2x} - \cancel{2} + \cancel{2x} + 2h}{h[(1-x)(1-(x+h))]}$$

$$= \lim_{h\to 0} \frac{2}{(1-x)(1-(x+h))} = \frac{2}{(1-x)(1-x)} = \frac{2}{(1-x)^2}$$

$$\therefore \frac{dy}{dx} = \frac{2}{(1-x)^2}$$

TECHNIQUES OF DERIVATIVES

1.) If $y = f(x)g(x)$ then $\frac{dy}{dx} = f'(x)g(x) + f(x)g'(x)$ Product rule

2.) If $y = \frac{f(x)}{g(x)}$ then $\frac{dy}{dx} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$ Quotient rule

3.) If $y = [f(x)]^n$ then $\frac{dy}{dx} = n[f(x)]^{n-1} f'(x)$ Power rule

4.) If $y = f(g(x))$ then $\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$ Chain rule

5.) If $y = f(t)$ then $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ Parametric defferentiation

$x = g(t)$

DERIVATIVES OF STANDARD FUNCTIONS

"$y = f(x)$"	"$y' = dy/dx$"
① $y = x^n$	$y' = n x^{n-1}$
② $y = e^{ax}$	$y' = a e^{ax}$
③ $y = a^x$	$y' = (\ln a) \cdot a^x$
4) $y = \ln x$	$y' = \frac{1}{x}$
5) $y = \sin x$	$y' = \cos x$
6) $y = \cos x$	$y' = -\sin x$
7) $y = \tan x$	$y' = \sec^2 x$
8) $y = \csc x$	$y' = -\csc x \cot x$
9) $y = \sec x$	$y' = \sec x \tan x$
10) $y = \cot x$	$y' = -\csc^2 x$
11) $y = \sin^{-1}(x)$	$y' = \frac{1}{\sqrt{1-x^2}}$
12) $y = \cos^{-1}(x)$	$y' = \frac{-1}{\sqrt{1-x^2}}$
13) $y = \tan^{-1}(x)$	$y' = \frac{1}{1+x^2}$
14)	

DERIVATIVE OF FUNCTIONS OF F(x)

	$y = f(x)$	$y' = \frac{dy}{dx} = f'(x)$
1)	$y = [f(x)]^n$	$y' = n\,[f(x)]^{n-1}\,f'(x)$
2)	$y = \sqrt{f(x)}$	$y' = \frac{1}{2}\,[f(x)]^{-1/2}\,f'(x) = \frac{f'(x)}{2\sqrt{f(x)}}$
3)	$y = e^{f(x)}$	$y' = f'(x)\,e^{f(x)}$
4)	$y = a^{f(x)}$	$y' = f'(x)\,a^{f(x)}\,(\ln a)$
5)	$y = \ln[f(x)]$	$y' = \frac{f'(x)}{f(x)}$
6)	$y = \sin[f(x)]$	$y' = f'(x)\cos[f(x)]$
7)	$y = \cos[f(x)]$	$y' = -f'(x)\sin[f(x)]$
8)	$y = \tan[f(x)]$	$y' = f'(x)\sec^2[f(x)]$
9)	$y = \csc[f(x)]$	$y' = -f'(x)\csc[f(x)]\cot[f(x)]$
10)	$y = \sec[f(x)]$	$y' = f'(x)\sec[f(x)]\tan[f(x)]$
11)	$y = \cot[f(x)]$	$y' = -f'(x)\csc^2[f(x)]$
12)	$y = \sin^{-1}[f(x)]$	$y' = \frac{f'(x)}{\sqrt{1-[f(x)]^2}}$
13)	$y = \cos^{-1}[f(x)]$	$y' = \frac{-f'(x)}{\sqrt{1-[f(x)]^2}}$

Example 2

Differentiate the following functions

(a) $y = 5^x + e^{-3x} + \sec 2x$

(b) $y = \tan 2x + \ln x - \cot 3x$

(c) $y = \ln(1+3x^2) + \sqrt{e^x + \sin x}$

(d) $y = \sqrt{2x^3+1}\ \tan^{-1}x$

(e) $y = e^{x^2}\left[\frac{1}{x^2} + \sin^{-1}\left(\frac{x}{2}\right) + \ln(2-3x)\right]$

Solution

(a) $\frac{dy}{dx} = (\ln 5)\,5^x - 3e^{-3x} + 2\sec 2x \tan 2x$

(b) $\frac{dy}{dx} = 2\sec^2 2x + \frac{1}{x} + 3\csc^2 3x$

(c) $\frac{dy}{dx} = \frac{6x}{1+3x^2} + \frac{e^x + \cos x}{2\sqrt{e^x + \sin x}}$

(d) $\frac{dy}{dx} = \frac{6x^2}{2\sqrt{2x^3+1}}\tan^{-1}x + \sqrt{2x^3+1}\cdot\left(\frac{1}{1+x^2}\right)$

$$= \frac{3x^2}{\sqrt{2x^3+1}}\tan^{-1}x + \frac{\sqrt{2x^3+1}}{1+x^2}$$

(e) $\frac{dy}{dx} = 2xe^{x^2}\left[\frac{1}{x^2} + \sin^{-1}\left(\frac{x}{2}\right) + \ln(2-3x)\right] + e^{x^2}\left[-\frac{2}{x^3} + \frac{\frac{1}{2}}{\sqrt{1-\left(\frac{x}{2}\right)^2}} - \frac{3}{2-3x}\right]$

$$\frac{dy}{dx} = 2xe^{x^2}\left[\frac{1}{x^2} + \sin^{-1}\left(\frac{x}{2}\right) + \ln(2-3x)\right] + e^{x^2}\left[\frac{-2}{x^3} + \frac{1}{\sqrt{4-x^2}} + \frac{3}{3x-2}\right]$$

Example 3

Find the first and second derivatives of

(a) $y = \sin t - \frac{1}{2}\cos 2t$, $x = 1 + \sin t$

(b) $y = 2\tan\theta$, $x = 3\sec\theta$ at $\theta = 45^\circ$

Solution (Parametric differentiation)

(a) $\frac{dy}{dt} = \cos t + \frac{1}{2}(2\sin 2t) = \cos t + \sin 2t$

Note: $\sin 2t = 2\cos t \sin t$

$\frac{dx}{dt} = \cos t$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\cos t + \sin 2t}{\cos t} = 1 + 2\sin t$$

$$\therefore \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt}(1 + 2\sin t)}{\cos t} = \frac{2\cos t}{\cos t} = 2$$

(b) We have: $\frac{dy}{d\theta} = 2\sec^2\theta$, $\frac{dx}{d\theta} = 3\sec\theta\tan\theta$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{2\sec^2\theta}{3\sec\theta\tan\theta} = \frac{2\sec\theta}{3\tan\theta} = \frac{2}{3}\csc\theta$$

$\theta = 45^\circ$

$\csc\theta = \frac{1}{\sin\theta} = \sqrt{2}$

At $\theta = 45^\circ$, $\left(\frac{dy}{dx}\right) = \frac{2}{3}\left(\frac{1}{\sin 45^\circ}\right) = \frac{2}{3}\sqrt{2}$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{d\theta}\left(\frac{dy}{dx}\right)}{\frac{dx}{d\theta}} = \frac{\frac{d}{d\theta}\left(\frac{2}{3}\csc\theta\right)}{3\sec\theta\tan\theta} = \frac{\frac{2}{3}(-\csc\theta\cot\theta)}{3\sec\theta\tan\theta} = -\frac{2}{9}\frac{\cos^3\theta}{\sin^3\theta}$$

At $\theta = 45^\circ$, $\left(\frac{d^2y}{dx^2}\right)_{\theta = 45^\circ} = -\frac{2}{9}\frac{\cos^3(45^\circ)}{\sin^3(45^\circ)} = -\frac{2}{9}$

$\cos(45^\circ) = \sin(45^\circ) = \frac{1}{\sqrt{2}}$

Example 4

Differentiate

(a) $y = \tan^{-1}\left[\dfrac{1+\tan x}{1-\tan x}\right]$ (b) $y = \dfrac{\sin^{-1}x}{\sqrt{1-x^2}}$

Solution

(a) Let $f(x) = \dfrac{1+\tan x}{1-\tan x}$, $f'(x) = \dfrac{\sec^2 x(1-\tan x)+\sec^2 x(1+\tan x)}{(1-\tan x)^2}$

$$f'(x) = \frac{2\sec^2 x}{(1-\tan x)^2}$$

$$\frac{dy}{dx} = \frac{f'(x)}{1+[f(x)]^2} = \frac{\dfrac{2\sec^2 x}{(1-\tan x)^2}}{1+\left(\dfrac{1+\tan x}{1-\tan x}\right)^2} = \frac{2\sec^2 x}{(1-\tan x)^2+(1+\tan x)^2}$$

$$\therefore \frac{dy}{dx} = \frac{\sec^2 x}{1+\tan^2 x} = \underline{\underline{1}}$$

Since: $\sec^2 x = 1+\tan^2 x$

(b) $$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}\left(\frac{1}{\sqrt{1-x^2}}\right) + \sin^{-1}x \cdot \left(-\frac{1}{2}(1-x^2)^{-3/2}\cdot(-2x)\right)$$

$$= \frac{1}{1-x^2} + \frac{x\sin^{-1}x}{(1-x^2)^{3/2}}$$

$$= \frac{1}{(1-x^2)}\left[1+\frac{x\sin^{-1}x}{\sqrt{1-x^2}}\right]$$

Example 5

Differentiate the following

(a) $y = \dfrac{e^{3x^2}\sin x}{\sqrt{2+x^3}}$ (b) $y = \left(\dfrac{5^x \cos 2x}{3+x^2}\right)^{2/3}$ (c) $y =$

Solution (Apply logarithmic differentiation)

(a) Step 1: (take log)

$$\ln y = \ln\left(e^{3x^2}\right) + \log(\sin x) - \ln(2+x^3)^{1/2}$$

$$\ln y = 3x^2 + \ln(\sin x) - \frac{1}{2}\ln(2+x^3)$$

step 2: (differentiate)

$$\frac{1}{y}\frac{dy}{dx} = 6x + \frac{\cos x}{\sin x} - \frac{1}{2}\cdot\left(\frac{3x^2}{2+x^3}\right)$$

$$\frac{dy}{dx} = y\left[6x + \cot x - \frac{3x^2}{2(2+x^3)}\right]$$

Since $y = \dfrac{e^{3x^2}\sin x}{\sqrt{2+x^3}}$

$$\therefore \frac{dy}{dx} = \frac{e^{3x^2}\sin x}{\sqrt{2+x^3}}\left[6x + \cot x - \frac{3x^2}{2(2+x^3)}\right]$$

(b) $$\ln y = \frac{2}{3}\left[x\ln 5 + \ln(\cos 2x) - \ln(3+x^2)\right]$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{2}{3}\left[\ln 5 - 2\tan 2x - \frac{2x}{3+x^2}\right]$$

Check !!

$$\therefore \frac{dy}{dx} = \frac{2}{3}\left(\frac{5^x\cos 2x}{3+x^2}\right)^{2/3}\left[\ln 5 - 2\tan 2x - \frac{2x}{3+x^2}\right]$$